Supplementary Material

Machine-learning analysis of blood metals and metabolic syndrome in NHANES 2007–2018

Table S1. Predictor dictionary. Each NHANES item code is given with a plain description of what the item records, its coding in this analysis, and the number of model columns it contributes after encoding (total 32). Categorical variables with more than two levels were one-hot encoded with the stated reference category omitted.

Domain	NHANES code	What the item records	Coding	Cols
Blood metals	LBXBPB	Blood lead concentration	Continuous, µg/dL	1
	LBXBCD	Blood cadmium concentration	Continuous, µg/L	1
	LBXTHG	Blood total mercury concentration	Continuous, µg/L	1
	LBXBSE	Blood selenium concentration	Continuous, µg/L	1
	LBXBMN	Blood manganese concentration	Continuous, µg/L	1
Demographic and socioeconomic	RIDAGEYR	Age at screening	Continuous, years	1
	RIAGENDR	Sex	Binary (female = 1)	1
	RIDRETH1	Race/Hispanic origin	One-hot; Mexican American = reference	5
	DMDEDUC2	Highest education level completed (adults)	Ordinal 1–5 (<9th grade → college graduate)	1
	DMDMARTL	Marital status	One-hot; married = reference	6
	DMDHHSIZ	Total number of people in the household	Continuous, count	1
	INDFMPIR	Ratio of family income to the federal poverty threshold	Continuous, 0–5	1
	RIDEXMON	Six-month period in which the examination took place	Binary (May–October = 1)	1
Anthropometric	BMXBMI	Body mass index from measured height and weight	Continuous, kg/m²	1
Lifestyle	SMQ020	Smoked at least 100 cigarettes in life	Binary (yes = 1)	1
	SMQ040	Current cigarette smoking frequency (never-smokers, who skip the item, coded not at all)	Ordinal, recoded 0 = not at all, 1 = some days, 2 = every day	1
	ALQ120Q / ALQ121	Any alcohol in the past 12 months, from the past-year drinking frequency (ALQ120Q in 2007–2016, ALQ121 in 2017–2018); those who never had 12 drinks in their life or never had a drink skip it and are coded no	Binary (yes = 1)	1
	ALQ130	Average number of alcoholic drinks on drinking days, past 12 months (0 for past-year non-drinkers)	Continuous, drinks/day	1

Domain	NHANES code	What the item records	Coding	Cols
	DBQ700	Self-rated overall diet quality	Ordinal 1–5 (excellent → poor)	1
	PAQ605	Vigorous-intensity activity as part of work	Binary (yes = 1)	1
	PAQ650	Vigorous-intensity recreational activity	Binary (yes = 1)	1
	PAQ665	Moderate-intensity recreational activity	Binary (yes = 1)	1
	PAD645	Minutes spent walking or bicycling for transportation on a typical day (gated on PAQ635, "walk or use a bicycle for transportation")	Continuous, minutes/day	1
	BPXPLS	Resting pulse, 60-second count	Continuous, beats/min	1

Survey modules used, all merged on the respondent sequence number (SEQN): demographics (DEMO), body measures (BMX), blood pressure (BPX), triglycerides (TRIGLY), HDL cholesterol (HDL), plasma fasting glucose (GLU), blood lead–cadmium–mercury–selenium–manganese (PBCD), alcohol use (ALQ), diet behavior (DBQ), smoking (SMQ), diabetes (DIQ), physical activity (PAQ), and blood pressure and cholesterol (BPQ). Hip circumference (BMXHIP) and physician-diagnosed diabetes (DIQ010) were screened but not retained in the primary predictor set.

Table S2. Metabolic syndrome component definitions. A component is scored positive when either the measured criterion or the drug-treatment criterion is met, following the harmonized joint interim statement [5], which retains the AHA/NHLBI thresholds and treatment clauses [6]. Participants meeting at least three of five components were classified as cases.

Component	Measured criterion	NHANES variable	Drug-treatment criterion
Abdominal obesity	Waist circumference ≥ 102 cm (men) or ≥ 88 cm (women)	BMXWAIST	Not applicable
Elevated triglycerides	Triglycerides ≥ 150 mg/dL	LBXTR	BPQ100D (currently taking prescribed medicine for cholesterol)
Reduced HDL cholesterol	HDL-C < 40 mg/dL (men) or < 50 mg/dL (women)	LBDHDD	BPQ100D (currently taking prescribed medicine for cholesterol)
Elevated blood pressure	Mean systolic ≥ 130 mmHg or mean diastolic ≥ 85 mmHg	BPXSY1–4, BPXDI1–4 (mean of valid readings)	BPQ050A (currently taking prescribed medicine for hypertension)
Elevated fasting glucose	Fasting plasma glucose ≥ 100 mg/dL	LBXGLU	DIQ050 (currently taking insulin); DIQ070 (taking pills to lower blood sugar)

HDL-C, high-density lipoprotein cholesterol. NHANES carries only one cholesterol-medication item and does not name the drug, whereas the guideline's lipid treatment clause refers to drugs such as fibrates and nicotinic acid; BPQ100D is applied to both lipid components as a stated modelling choice, and a sensitivity analysis without it is given in Table S7. Blood pressure is the mean of all valid readings (a diastolic reading of 0 mmHg is treated as invalid); a first-reading-only version is also tested in Table S7. Component measurements and medication items were used solely to construct the outcome and were excluded from the predictor set.

Table S3. Hyperparameter search space, selected configuration and cross-validated discrimination for each algorithm. Every grid holds at most 12 configurations and was searched exhaustively, with 10-fold stratified cross-validation on the training partition, optimizing AUROC. *These values are tuning-selection statistics and are optimistically biased; test-partition performance is reported in the main text.*

Algorithm	Search space	Selected configuration	CV AUROC
CatBoost	iterations ∈ {200, 500}; learning_rate ∈ {0.01, 0.05, 0.1}; depth ∈ {4, 6}	learning_rate = 0.05, iterations = 200, depth = 6	0.8438
LightGBM	n_estimators ∈ {200, 500}; learning_rate ∈ {0.01, 0.05, 0.1}; num_leaves ∈ {15, 31}	num_leaves = 15, n_estimators = 500, learning_rate = 0.01	0.8400
XGBoost	n_estimators ∈ {200, 500}; learning_rate ∈ {0.01, 0.05, 0.1}; max_depth ∈ {3, 5}	n_estimators = 200, max_depth = 3, learning_rate = 0.05	0.8399
Gradient Boosting	n_estimators ∈ {100, 200}; learning_rate ∈ {0.01, 0.1}; max_depth ∈ {2, 3}	n_estimators = 100, max_depth = 3, learning_rate = 0.1	0.8393
SVM	C ∈ {0.1, 1, 10}; kernel ∈ {RBF, linear}	kernel = rbf, C = 1	0.8374
Random Forest	n_estimators ∈ {200, 500}; max_depth ∈ {None, 8, 15}; min_samples_leaf ∈ {1, 5}	n_estimators = 500, min_samples_leaf = 5, max_depth = None	0.8371
Logistic Regression	C ∈ {0.01, 0.1, 1, 10}; penalty = L2	penalty = l2, C = 0.01	0.8352
Decision Tree	max_depth ∈ {3, 5, 8, None}; min_samples_leaf ∈ {1, 5, 20}	min_samples_leaf = 20, max_depth = 5	0.8184
KNN	n_neighbors ∈ {5, 15, 25}	n_neighbors = 25	0.8018
MLP	hidden_layer_sizes ∈ {(64), (64, 32)}; alpha ∈ {0.0001, 0.001}	hidden_layer_sizes = (64), alpha = 0.001	0.7898
Naive Bayes	none (no hyperparameters searched)	–	0.7438

Table S4. Classification performance on the held-out test partition at the decision threshold selected by maximizing the Youden index on out-of-fold training predictions, and at the conventional 0.5 cut-point for comparison. All values except the threshold, F1 and κ are percentages.

Model	Thresh.	Acc.	Sens.	Spec.	PPV	NPV	F1	κ	$Acc._{0.5}$	$Sens._{0.5}$	$Spec._{0.5}$
CatBoost	0.511	77.3	78.3	76.4	74.4	80.1	0.763	0.545	77.2	79.4	75.3
XGBoost	0.531	76.8	76.2	77.2	74.6	78.7	0.754	0.533	76.7	79.3	74.4
Gradient Boosting	0.552	76.4	74.0	78.5	75.0	77.5	0.745	0.525	76.5	78.3	74.9
LightGBM	0.532	76.6	75.0	78.0	74.9	78.1	0.750	0.530	76.4	77.8	75.1
Random Forest	0.486	76.4	78.8	74.3	72.9	80.0	0.757	0.529	76.3	77.3	75.4
Logistic Regression	0.446	76.8	83.3	71.1	71.6	82.9	0.770	0.538	76.5	76.3	76.6
SVM	0.473	76.0	79.2	73.1	72.1	80.1	0.755	0.520	76.3	77.8	74.9
Decision Tree	0.493	75.0	72.8	77.0	73.5	76.4	0.732	0.498	75.0	72.8	77.0
KNN	0.520	71.7	77.5	66.7	67.1	77.2	0.719	0.438	71.7	77.5	66.7
MLP	0.338	72.8	82.7	64.2	66.9	80.9	0.740	0.462	74.0	74.5	73.5
Naive Bayes	0.220	67.4	74.3	61.3	62.7	73.2	0.680	0.352	68.5	64.2	72.3

Acc., accuracy; Sens., sensitivity; Spec., specificity; PPV, positive predictive value; NPV, negative predictive value; κ, Cohen's kappa. Subscript 0.5 denotes the conventional cut-point. Thresholds were selected without reference to the test partition. Bootstrap 95% confidence intervals for every value are given in 04_Results_Tables/model_comparison.csv.

Table S5. The fifteen highest-ranked predictors by mean absolute SHAP value, with their permutation-importance values and ranks for comparison. Both methods are applied to the CatBoost model on the held-out test partition and both rank over the same 32 predictors.

Predictor	Mean |SHAP|	SHAP rank	Permutation Δ AUROC (mean ± SD)	Perm. rank	Above noise
Body mass index (kg/m²)	1.0474	1	0.16527 ± 0.00795	1	yes
Age (years)	0.9382	2	0.14661 ± 0.01000	2	yes
Vigorous recreational activity	0.1753	3	0.00303 ± 0.00110	4	yes
Resting pulse (60 sec)	0.1516	4	0.00548 ± 0.00167	3	yes
Self-rated diet quality	0.1016	5	0.00070 ± 0.00071	14	no
Education level	0.0773	6	0.00009 ± 0.00051	21	no
Race/ethnicity: Non-Hispanic Black	0.0629	7	0.00134 ± 0.00058	9	yes
Avg. drinks per drinking day (past 12 mo; 0 = non-drinker)	0.0562	8	0.00178 ± 0.00119	6	yes
Marital status: Never married	0.0561	9	0.00105 ± 0.00039	12	yes
Sex: Female	0.0446	10	0.00271 ± 0.00065	5	yes
Current smoking intensity	0.0445	11	0.00120 ± 0.00063	10	yes
Household size	0.0407	12	−0.00046 ± 0.00032	31	yes
Race/ethnicity: Other/Multi-Racial	0.0368	13	0.00151 ± 0.00040	7	yes
Minutes walking/biking for transportation	0.0363	14	0.00009 ± 0.00059	20	no
Blood cadmium (µg/L)	0.0311	15	0.00100 ± 0.00042	13	yes

Permutation Δ AUROC is the mean decrease in test-set AUROC over ten permutations of the predictor, with the standard deviation across those permutations. *Above noise* is a descriptive heuristic, not a hypothesis test: it flags predictors whose mean exceeds one permutation standard deviation, a roughly one-sigma bar reflecting permutation variability only and not sampling variability of the test partition. Only 12 of these fifteen clear it, so ranks below the leading few should not be read as a resolved ordering. A negative value means permuting the predictor left discrimination unchanged or marginally improved it; 5 of the 32 predictors had a negative mean and 1 exactly zero, so the lower permutation ranks are not interpretable.

Table S6. Discrimination and calibration of the CatBoost model within subgroups of the held-out test partition. Calibration is reported alongside discrimination because a model can rank participants poorly within a stratum while still stating their absolute risk correctly, and the two have different consequences in use.

Subgroup	n	Prev. (%)	AUROC (95% CI)	Cal. slope (95% CI)	O:E ratio (95% CI)
Overall	2,176	46.7	0.853 (0.837–0.869)	1.06 (0.97–1.16)	0.983 (0.945–1.016)
Male	1,031	48.3	0.858 (0.834–0.880)	1.09 (0.97–1.25)	1.031 (0.980–1.082)
Female	1,145	45.2	0.851 (0.829–0.872)	1.04 (0.94–1.18)	0.940 (0.892–0.989)
Age 20–40 y	760	22.9	0.855 (0.824–0.887)	1.03 (0.87–1.24)	0.924 (0.827–1.025)
Age 41–59 y	717	48.5	0.795 (0.764–0.828)	1.02 (0.88–1.21)	0.989 (0.923–1.051)
Age > 59 y	699	70.7	0.780 (0.740–0.816)	1.09 (0.90–1.31)	1.001 (0.959–1.044)
Non-Hispanic White	908	49.1	0.850 (0.826–0.873)	1.02 (0.90–1.15)	0.974 (0.921–1.026)
Mexican American	337	45.1	0.830 (0.783–0.871)	1.02 (0.79–1.33)	0.962 (0.861–1.052)
Non-Hispanic Black	424	46.9	0.861 (0.828–0.895)	1.17 (0.97–1.46)	0.990 (0.913–1.076)
Other Hispanic	233	47.2	0.858 (0.807–0.903)	1.13 (0.88–1.56)	0.966 (0.859–1.067)
Other/Multi-racial	274	39.8	0.877 (0.834–0.915)	1.07 (0.82–1.43)	1.060 (0.935–1.177)

Cal. slope, calibration slope, describing dispersion of the predictions (1.00 = perfect); O:E, ratio of observed to expected outcome frequency, describing calibration in the large (1.00 = perfect). All intervals come from the same 1,000 percentile bootstrap resamples, with the calibration model refitted in each. Slope and O:E answer different questions and are not interchangeable. Age groups are tertiles of the test-partition age distribution. Subgroups are nested within the overall sample, so their intervals are not independent of the overall row and should not be used to test against it.

Table S7. Sensitivity analyses. Each entry is the mean AUROC from 10-fold stratified cross-validation on the training partition, using the three leading algorithms at their tuned hyperparameters (Table S3), holding everything except the stated change fixed. The outcome-definition arms change the label, and therefore its prevalence, as well as the difficulty of the task.

Analysis	Model	Primary	Alternative	Δ AUROC
Add diagnosed diabetes	CatBoost	0.8438	0.8533	+0.0095
Add diagnosed diabetes	LightGBM	0.8400	0.8504	+0.0104
Add diagnosed diabetes	XGBoost	0.8399	0.8512	+0.0112
Class weighting instead of SMOTENC	CatBoost	0.8438	0.8430	-0.0008
Class weighting instead of SMOTENC	LightGBM	0.8400	0.8387	-0.0013
Class weighting instead of SMOTENC	XGBoost	0.8399	0.8397	-0.0003
Lead only, other four metals dropped	CatBoost	0.8438	0.8428	-0.0010
Lead only, other four metals dropped	LightGBM	0.8400	0.8401	+0.0002
Lead only, other four metals dropped	XGBoost	0.8399	0.8399	-0.0000
Measured-value-only outcome definition	CatBoost	0.8438	0.8067	-0.0371
No lipid-medication clause in the outcome	CatBoost	0.8438	0.8381	-0.0057
Outcome with first blood-pressure reading only	CatBoost	0.8438	0.8436	-0.0002
Measured-value-only outcome definition	LightGBM	0.8400	0.8044	-0.0356
No lipid-medication clause in the outcome	LightGBM	0.8400	0.8353	-0.0047

Analysis	Model	Primary	Alternative	Δ AUROC
Outcome with first blood-pressure reading only	LightGBM	0.8400	0.8408	+0.0008
Measured-value-only outcome definition	XGBoost	0.8399	0.8025	-0.0374
No lipid-medication clause in the outcome	XGBoost	0.8399	0.8347	-0.0052
Outcome with first blood-pressure reading only	XGBoost	0.8399	0.8418	+0.0019

Δ AUROC is the alternative specification minus the primary, so a positive value indicates the alternative discriminated better. For the outcome-definition arms the primary specification is the harmonized definition used throughout. Gradient boosting has no class-weight option and used balanced sample weights instead.

Table S8. Incremental value of the blood metals and simple baselines. The leading algorithm (CatBoost) was refitted with its tuned hyperparameters on reduced predictor sets; every model was scored on the same 2,176 test participants and compared with the full model by DeLong's test, with a paired-bootstrap 95% CI for the difference.

Model	Columns	CV AUROC	Test AUROC (95% CI)	Δ vs full (95% CI)	p
Full model (all predictors)	32	0.8438	0.853 (0.837–0.869)	reference	–
Without the five metals	27	0.8423	0.853 (0.837–0.869)	−0.0001 (−0.0028 to 0.0028)	0.951
Lead as the only metal	28	0.8428	0.853 (0.837–0.869)	−0.0004 (−0.0031 to 0.0023)	0.778
Without body mass index	31	0.7734	0.774 (0.751–0.793)	−0.0796 (−0.0949 to −0.0641)	< 0.001
Age and body mass index only (logistic regression)	2	0.8264	0.835 (0.817–0.853)	−0.0186 (−0.0263 to −0.0113)	< 0.001

CV AUROC, mean of 10-fold cross-validation on the training partition. Removing body mass index also removes its missing-value indicator. Waist circumference defines one outcome component and correlates strongly with body mass index, so part of the no-BMI loss is definitional.

Table S9. Repeated stratified cross-validation on the training partition (10 folds × 5 repeats, 50 fits per model, tuned hyperparameters), showing how much discrimination varies between data partitions.

Model	Mean AUROC	SD across folds	2.5th–97.5th percentile	Range of repeat means
CatBoost	0.8419	0.0116	0.8219–0.8627	0.8397–0.8438
XGBoost	0.8401	0.0112	0.8225–0.8613	0.8388–0.8418
LightGBM	0.8397	0.0114	0.8183–0.8610	0.8383–0.8411
Gradient Boosting	0.8392	0.0110	0.8195–0.8603	0.8382–0.8405
SVM	0.8372	0.0115	0.8137–0.8595	0.8352–0.8384
Random Forest	0.8366	0.0118	0.8162–0.8582	0.8351–0.8375
Logistic Regression	0.8356	0.0123	0.8131–0.8609	0.8352–0.8362
Decision Tree	0.8177	0.0125	0.7923–0.8396	0.8154–0.8195
KNN	0.8014	0.0133	0.7708–0.8249	0.7999–0.8028
MLP	0.7896	0.0164	0.7584–0.8203	0.7880–0.7924
Naive Bayes	0.7438	0.0172	0.7165–0.7717	0.7429–0.7454

Table S10. Temporal validation of the CatBoost model: development on the 2007–2014 cycles with hyperparameters re-tuned by 10-fold cross-validation on those cycles only, testing on 2015–2018, compared with the primary random split.

Design	Train n	Test n	AUROC (95% CI)	Cal. slope	Cal. intercept	O:E
Random split (primary analysis)	8,703	2,176	0.853 (0.837–0.869)	1.06	−0.051	0.983
Temporal holdout (develop 2007-2014, test 2015-2018; re-tuned)	7,786	3,093	0.833 (0.819–0.846)	1.06	−0.008	0.997

Cal. intercept, calibration-in-the-large (slope fixed at 1; 0 = perfect).

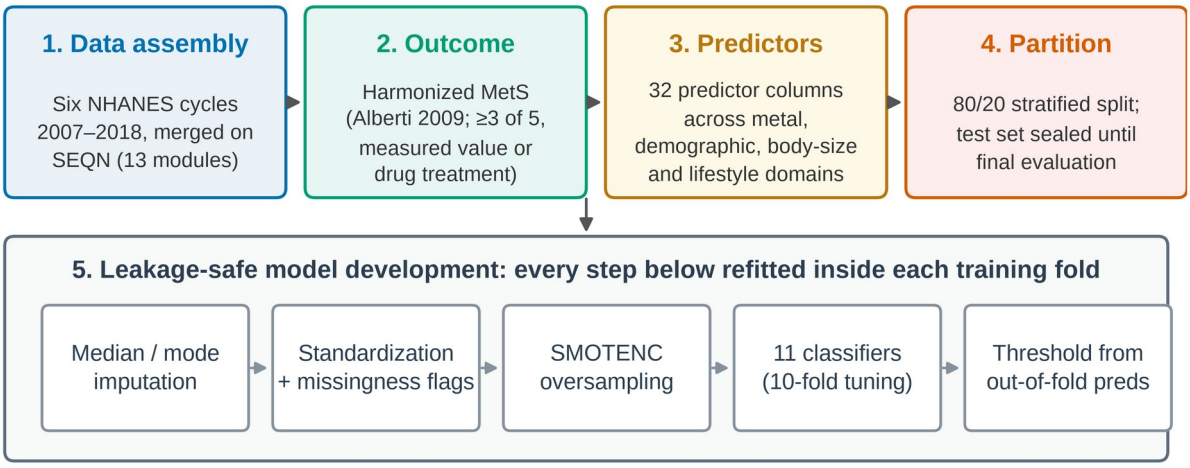


Figure S1. Analysis pipeline. Stages 1–4 are performed once on the cohort. Every step in stage 5 is refitted independently inside each training fold during cross-validation, and inside the training partition alone for the final models, so that no quantity estimated from held-out data can influence a model later scored on it.

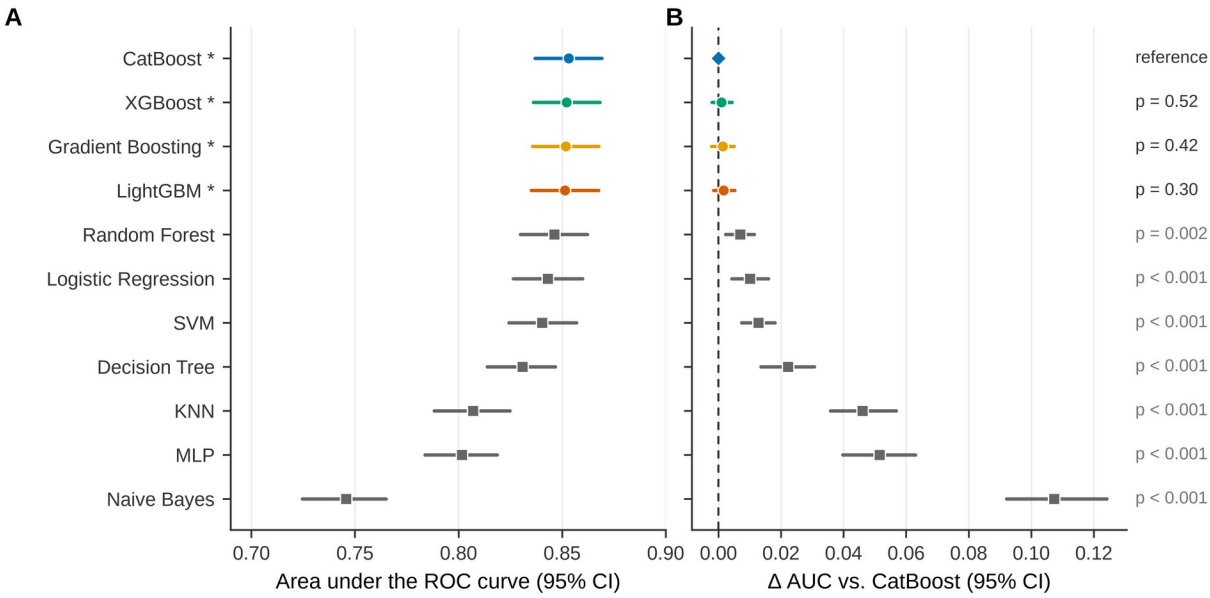


Figure S2. Model discrimination and pairwise comparison against CatBoost. **(A)** AUROC with bootstrap 95% confidence intervals. **(B)** Difference in AUROC from CatBoost, with a paired-bootstrap 95% confidence interval (the same resampled participants scored under both models) and the DeLong p-value; intervals crossing zero indicate models that cannot be distinguished from CatBoost. Using unpaired intervals from panel A for these comparisons would overstate the uncertainty, because all models are scored on the same participants.

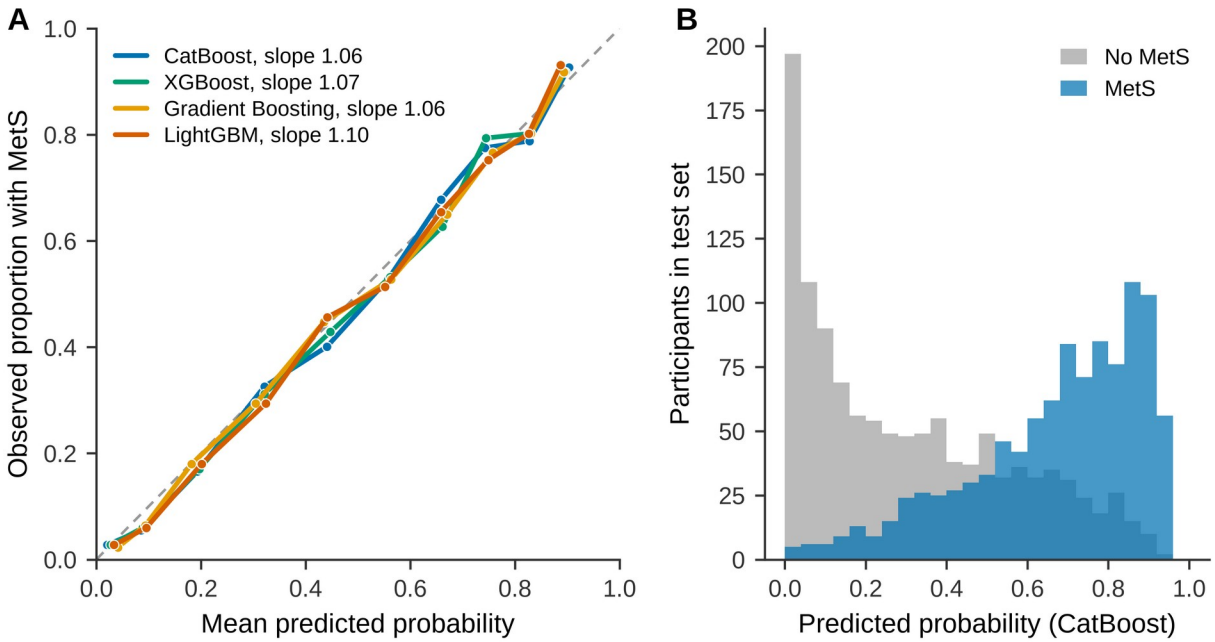


Figure S3. Calibration on the held-out test partition. **(A)** Calibration curves for the four leading models using ten quantile-based bins, with calibration slopes in the legend; the diagonal denotes perfect calibration. **(B)** Distribution of predicted probabilities from CatBoost by true outcome status.

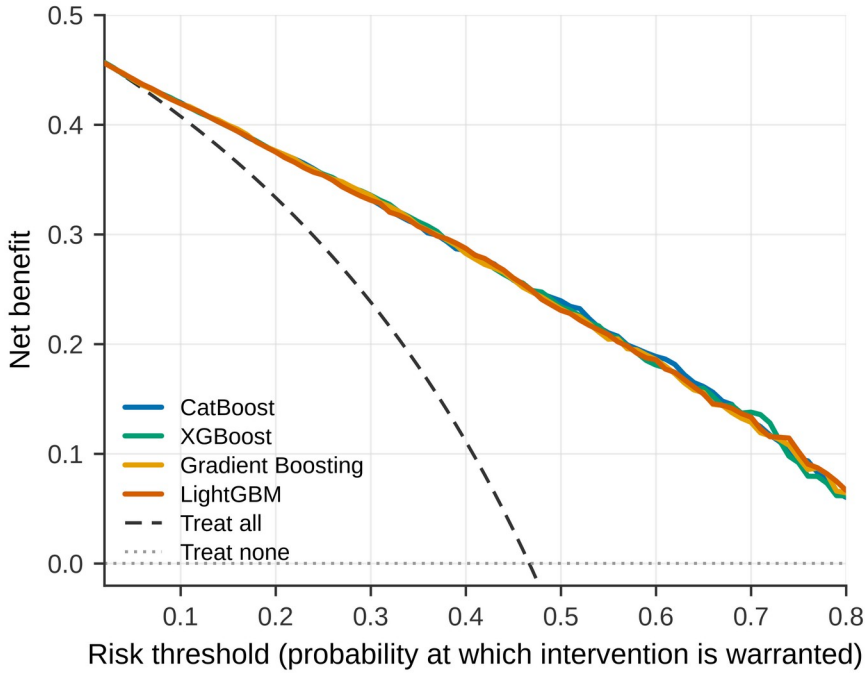


Figure S4. Decision-curve analysis on the held-out test partition. Net benefit is plotted against the risk threshold (2% to 80%) at which intervention would be considered warranted, for the four leading models and for the treat-all and treat-none reference strategies.

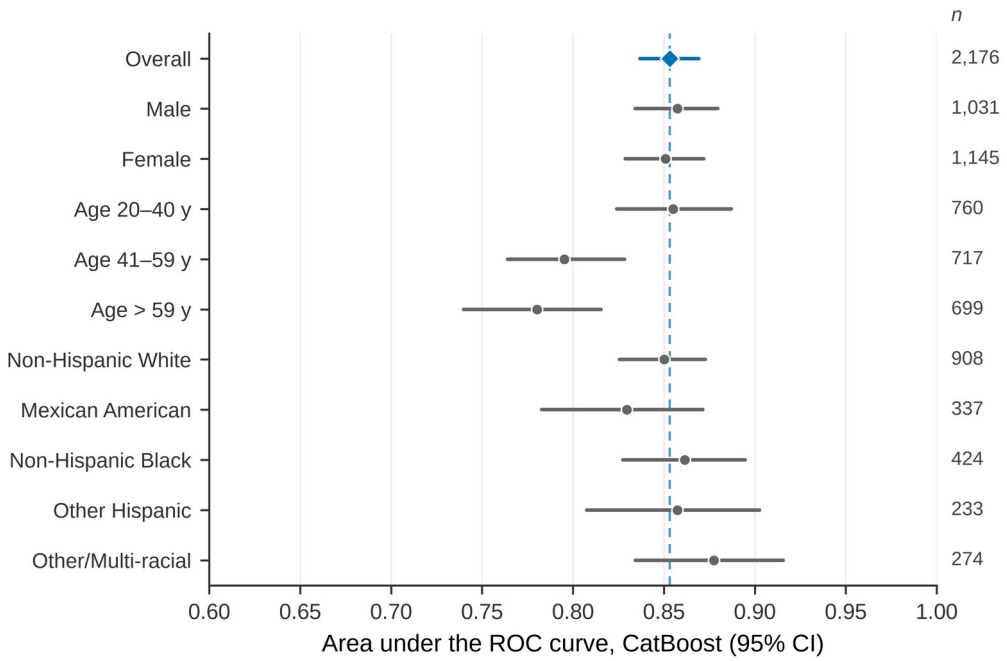


Figure S5. Discrimination of the CatBoost model within subgroups of the held-out test partition, with bootstrap 95% confidence intervals from 1,000 resamples. The dashed vertical line marks the overall test-set AUROC and subgroup sizes are given at the right. Age groups are tertiles of the test-partition age distribution. Because every subgroup is nested within the overall sample, these intervals are not independent of the overall estimate and overlap with it should not be read as a significance test. Subgroup calibration is reported in Table S6.